Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:03-01-2025
Time: 3HRS GTM-10 Max. Marks: 300

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	C	C	A	A	A	B	C	D	B
11	12	13	14	15	16	17	18	19	20
B	C	B	D	C	A	C	D	A	D
21	22	23	24	25					
4	500	200	21	200					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	B	D	A	C	C	D	C	C
36	37	38	39	40	41	42	43	44	45
D	B	C	B	A	D	D	D	A	A
46	47	48	49	50					
13	92	1200	272	15					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	A	D	C	D	B	D	A	C	B
61	62	63	64	65	66	67	68	69	70
D	B	B	B	D	A	B	A	B	C
71	72	73	74	75					
15	4	30	9	3					

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1. $\frac{1}{\lambda}=R\left(\frac{1}{2^2}-\frac{1}{\infty^2}\right)\Rightarrow\lambda=\frac{4}{1.1\times10^7}m=3636A^0$

2. Area of cycle gives work done

3. $U=\frac{1}{2}\varepsilon_0E^2V=\frac{1}{2}\times9\times10^{-12}\times\frac{10^4}{2}\times10^{-4}=2.25\times10^{-12}J$

4. $g_d=g_s\left(1-\frac{d}{R}\right)=\frac{3g_s}{4}$

5. $\lambda=\frac{12400}{E(in\ eV)}A^0=\frac{12400}{2.1}A^0$

6. Work done = surface tension × increase in area
$= 72 \times [10 \times 0.7 - 10 \times 0.5] \times 2 = 288$ erg

7. $V_{\max}=\sqrt{Rg\left(\frac{\tan\theta+\mu}{1-\mu\tan\theta}\right)}=\sqrt{100\times10\left(\frac{0.75+0.25}{1-0.75\times0.25}\right)}$

8. $F\propto Q^2$

$F^1\propto\left(\frac{Q}{2}\right)\left(\frac{Q}{4}\right)$

$\frac{64}{F^1}=8$

$F^1=8N$

9. $E=\frac{1}{2}LI^2$

10. $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}\Rightarrow\frac{1}{f}=\frac{1}{10}-\frac{1}{20}=\frac{1}{20}\Rightarrow f=20Cm$

11. Avg K.E of each molecule is $E_{av}=E=\frac{3}{2}K_BT$

If N_1 is the no. of molecules of hydrogen and N_2 is no. of molecules of oxygen, then

Total K.E of hydrogen $E_1=N_1\left(\frac{3}{2}K_BT\right)$

Total K.E of Oxygen is $E_2=N_2\left(\frac{3}{2}K_BT\right)$

$\Rightarrow\frac{E_1}{E_2}=\frac{N_1}{N_2}=2$

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12. Spring balance reading $= m(g+a)$

13. $\frac{hc}{\lambda} - \phi = eV_s \Rightarrow \frac{1240}{400} - \Phi = 1.5 \Rightarrow \Phi = 1.6eV$

14. $P = VI \Rightarrow p = V\left(\frac{ne}{t}\right) \Rightarrow \frac{n}{t} = \frac{p}{Ve} = \frac{100}{240 \times 1.6 \times 10^{-19}}$

15. $P_i = p \qquad k_i = k \qquad \Rightarrow \frac{\Delta k}{k} \times 100 = 800\%$

$p_f = 3p \qquad k_f = 9k$

16. $\frac{4}{3}\pi\left(R^1\right)^3 = 8 \times \frac{4}{3}\pi R^3 \Rightarrow R^1 = 2R$

17. 1) $\chi \propto \frac{1}{T}$ 2) $B = 0\ r < R, B \propto \frac{1}{r}\ r > R$

3) $B \propto r, r < R,\ B \propto \frac{1}{r}\ r > R$ 4) $B = \frac{\mu_0}{4\pi}\frac{I\pi R^2}{\left(R^2 + r^2\right)^{3/2}}$

18. Emf is induced in the disc and it opposes the motion. Hence due to the resistance of the ring all energy dissipates

19. $E = \frac{hc}{\lambda} \Rightarrow ML^2T^{-2} = [h]\left[\frac{LT^{-1}}{L}\right] \Rightarrow [h] = ML^2T^{-1}$

$B = \frac{\triangle p}{\triangle v / v} = \left[\frac{F}{A}\right] = ML^{-1}T^{-2}$

20. $\vec{S} = \vec{ut} + \frac{1}{2}\vec{at}^2$

21. $mg = qE \Rightarrow q = ne$

22. $V = \sqrt{(400)^2 + (600-300)^2} = 500v$

23. $\upsilon = t^2 - 1 \Rightarrow \frac{dx}{dt} = t^2 - 1 \Rightarrow dx = \left(t^2 - 1\right)dt$

$\Rightarrow x = \frac{t^3}{3} - t$

From 0 to 1s: $\Delta x = -\frac{2}{3}\text{m}$

From 1 to 2s: $\Delta x = \frac{4}{3}\text{m}$

Total distance = 2m

24. $I = mr^2 = (1+2+3+4+5+6) \times 1^2 = 21$

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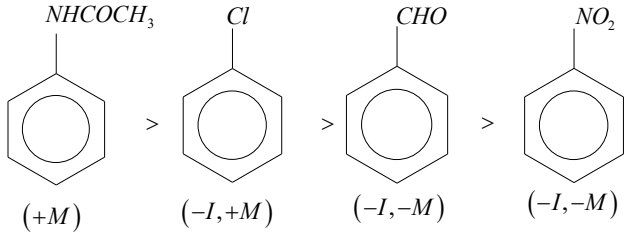
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25. $\Delta U = U_f - U_i = -\left(MB\cos 180^0\right) - \left(-MB\cos\theta^0\right) = 2MB$

$$\Delta U = 2\times 50\times \pi \times \left(\sqrt{\frac{7}{22}}\right)^2 \times 2\times 10^{-3} = 200mJ$$

CHEMISTRY

26.



27. Wt. of $H_2SO_4 = 93$ gm in 100 ml solution

Wt. of solution=vol $\times$density = 100ml $\times$1.84 gm =184 gm

Wt. of solvent = 184 – 93 = 91 gm

$$m = \frac{93}{98}\times\frac{1000}{91} = 10.428 \text{ m}$$

29. Conceptual

30. $Cl^-_{(aq)} \xrightarrow[H_2SO_4]{K_2Cr_2O_7} \underset{(\text{Reddish brown vapours})}{CrO_2Cl_2}\uparrow \xrightarrow{NaOH(aq)} \underset{(aq)}{Na_2CrO_4} \xrightarrow[CH_3COOH]{(CH_3COO)_2Pb} \underset{\text{Yellow ppt}}{PbCrO_4}\downarrow$

32. $Mn(Z=25),\ Mn^{+2} = 3d^5$

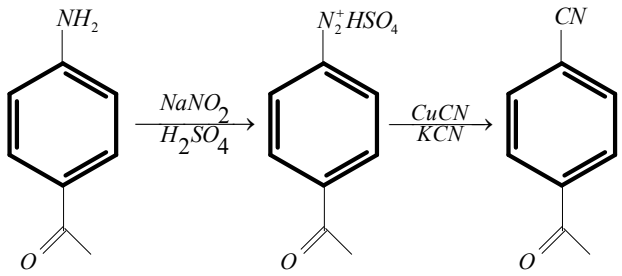
$Cr(Z=24),\ Cr^{+2} = 3d^4$

$V(Z=23),\ V^{+2} = 3d^3$

Since $\mu = \sqrt{n(n+2)}\ BM$ (n= no. of unpaired e^-)

Hence correct order :- $Mn^{+2} > Cr^{+2} > V^{+2}$

34.



(A) (B)

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35. Meso compounds are internally compensated from where as racemic mixtures are externally compensated modifications

36. $PF_3 \rightarrow$ Trigonal pyramidal shape with tetrahedral orientation of e pairs

$BrF_3 \rightarrow$ T- shaped with trigonal bipyramidal geometry

$XeF_4 \rightarrow sp^3d^2$

$[CoF_6]^{3-} \rightarrow sp^3d^2$

37. $P^H = P^{Ka} + \log\dfrac{[C_6H_5COO^-]}{[C_6H_5COOH]}$

$4.5 = 4.2 + \log\dfrac{[C_6H_5COO^-]}{[C_6H_5COOH]}$

$\log\dfrac{[C_6H_5COO^-]}{[C_6H_5COOH]} = 4.5 - 4.2 = 0.3$

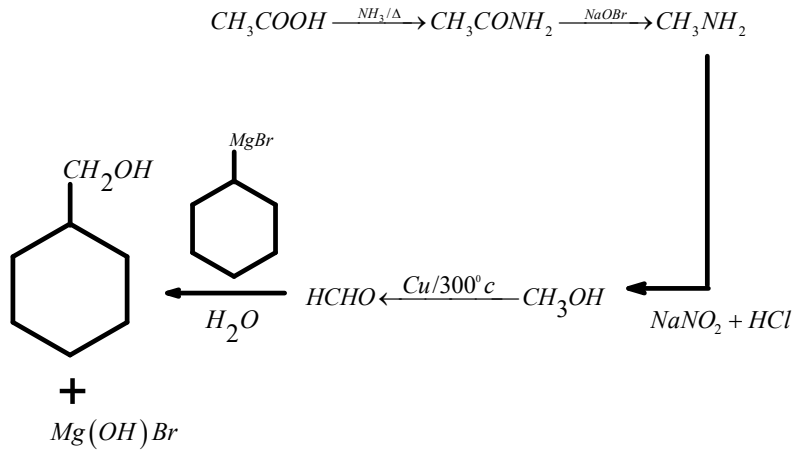
$\dfrac{[C_6H_5COO^-]}{[C_6H_5COOH]} = 2 = \dfrac{C_SV_S}{C_aV_a}$

But vol.of benzoic acid $= V_{ml}$

$2 = \dfrac{0.2 \times (300 - V)}{0.1 \times V}$

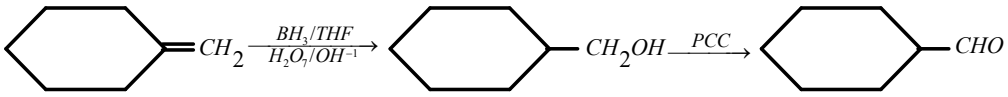
Thus $V_{ml} = 150ml$

38.



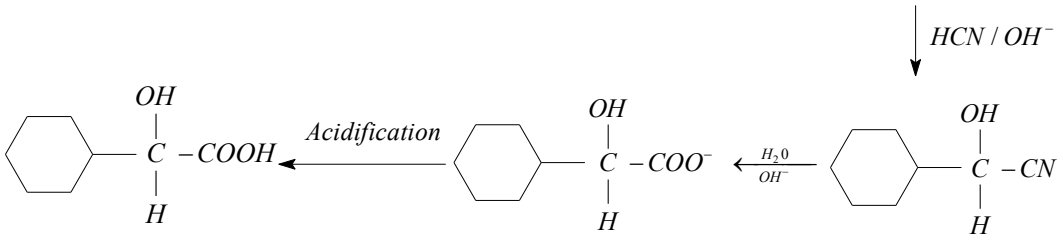
39. Conceptual

42.



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43. Conceptual

44. in octahedral complexes

i) if $\Delta_0 < p$ (weak field high spin situation) the fourth electron enters one of the e_g orbital giving the configuration $t_{2g}^3 e_g^1$

ii) if $\Delta_0 > p$ (strong field, low spin situation) pairing will occur in t_{2g} level as $t_{2g}^4 e_g^0$

45. $CO_{(g)} + Cl_{2_{(g)}} \rightleftharpoons COCl_{2_{(g)}}$

$\Delta n = n_p - n_R$

$= 1 - 2 = -1$

$k_p = k_c.(RT)^{\Delta n}$

$\therefore \frac{k_p}{k_c} = (RT)^{(-1)} = \frac{1}{(RT)}$

46. except $NH_{3(g)}$ *and* $H_2O_{(g)}$, All others have H-bonding

47. $2CH_3COOH \rightleftarrows (CH_3COOH)_2$

$1-\alpha \qquad \frac{\alpha}{2} \qquad \Rightarrow \left(i = 1 - \frac{\alpha}{2}\right)$

$\Delta T = i.k_t.m$

$0.45 = \left(1 - \frac{\alpha}{2}\right)(5.0)\left(\frac{0.2}{60} \times \frac{1000}{20}\right)$

$1 - \frac{\alpha}{2} = 0.54$

$\alpha = 0.92$

This percentage of association =92%

48. ΔS^0 for the reaction $\frac{1}{2}x_2 + \frac{3}{2}y_2 \rightleftharpoons xy_3$

$\Delta S^0 = 50 - \left[\frac{1}{2} \times 60 + \frac{3}{2} \times 30\right] = -25J$

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For equilibrium $\Delta G = 0$

$\Delta G = \Delta H - T\Delta S = 0$

$\Delta H = T\Delta S$

$T = \frac{\Delta H}{\Delta S} = \frac{-30000}{-25} = 1200K$

49. mvr $= \frac{nh}{2\pi} = 3.15 \times 10^{-34}$

So $n = 3$

Total energy of $e^- = -13.6\left(\frac{z^2}{n^2}\right) = -13.6\left(\frac{3^2}{3^2}\right) = -13.6ev$

P.E =2 T.E) =2(-13.6)

Thus p =-27.2 ev

$\therefore |p| \times 10^1 = 272$

50. Half-life for a zero order reaction is given by $t_{1/2} = \left[\frac{A_0}{2K}\right]$

$K_0 = \frac{[A_0]}{2t_{1/2}} = \frac{2.0\ mol/lit}{2 \times 1hr} = 1.0\ mol/lit/hr$

$K_0 = \frac{1}{t}\left[(A_0) - (A)\right]$

$t = \frac{1}{k}\left[(A_0) - (A)\right]$

$t = \frac{(0.5 - 0.25)mol/lit}{1mol/lit/hr} = 0.25hr = 15\min$

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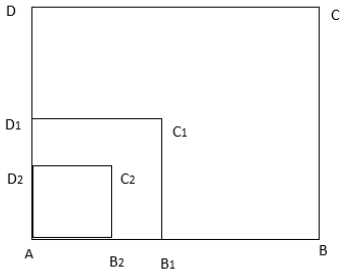
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51. Let side of square $\square ABCD$ be 's'

Area of $\square ABCD = S^2$

Area of $\square AB_1C_1D_1 = \frac{S^2}{4}$

Area of $\square AB_2C_2D_2 = \frac{S^2}{16}$ etc



$\therefore$ sum of Areas

$= S^2 + \frac{S^2}{4} + \frac{S^2}{16} + - - - = \frac{4S^2}{3} = 12$

$S^2 = 9$ & S=3

52. $R = \{(2,2)(2,4),(2,6),(2,8),(4,4),(4,6),(4,2),(6,2),(6,4),(8,2)\}$

$\Rightarrow m = 10$

To make it equivalent we need to add

(6,6),(8,8),(8,6),(4,8),(8,4),(6,8)

$n = 6$

$so \Rightarrow mn = 10 \times 6 = 60$

53. $n(s) = \frac{10!}{2!3!}$

$n(\in) = \frac{8!}{2!} - 7!$

$\therefore \text{Probability} = \frac{8! - 7!(2)}{(2)(10!)} \times 2!3!$

$= \frac{(8-2)}{8 \times 9 \times 10} \times 6$

$= \frac{1}{20}$

54. Clearly we have

$\frac{xdy + ydx}{e^{xy}} = \frac{xdy - ydx}{x^2}$

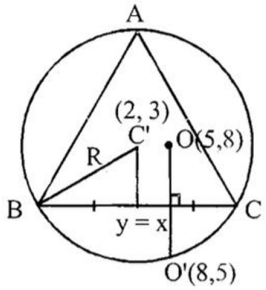
$\Rightarrow \int d(xy)e^{-xy} = \int d\left(\frac{y}{x}\right) \Rightarrow \frac{y}{x} + e^{-xy} = c$

55. Image of orthocentre $(5,8)$ in the base BC $y = x$ is $(8,5)$ which lies on circle.

Hence radius of the circle is $O'C' = \sqrt{40}$

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$\Rightarrow$ equation of circle is $(x-2)^2+(y-3)^2=40$

$\Rightarrow\ x^2+y^2-4x-6y-27=0$

56. On simplification $\underset{n\to\infty}{Lt}\dfrac{\left(\prod_{K=0}^{n+1}{}^{n+1}C_K\right)\left(\prod_{K=0}^{n-1}{}^{n-1}C_K\right)}{\left(\prod_{K=0}^{n}{}^{n}C_K\right)^2}$

$=\underset{n\to\infty}{Lt}\left(\dfrac{n+1}{n}\right)^n=e$

57. $\left(1-2\sqrt{x}\right)^{20}=1-{}^{20}C_1\left(2\sqrt{x}\right)+{}^{20}C_2\left(2\sqrt{x}\right)^2+\ldots$

$\left(1-2\sqrt{x}\right)^{20}=1+{}^{20}C_1\left(2\sqrt{x}\right)+{}^{20}C_2\left(2\sqrt{x}\right)^2+\ldots$

Adding

$=2\left(1+{}^{20}C_2.2^2.x+{}^{20}C_4.2^4.x^2+\ldots\right)$

Put x=1 or required sum

$=\dfrac{(1-2)^{20}+(1+2)^{20}}{2}$

$=\dfrac{3^{20}+1}{2}$

58. If α,β are the roots, then

$a^2+b^2=\left(\alpha^2+1\right)\left(\beta^2+1\right)$

59. We have $w=z+\dfrac{1}{z}$

Take $z=e^{i\theta}$ ($\because |z|=1$).

We have $w=e^{i\theta}+e^{-i\theta}=2\cos\theta$

We know that w is real and $|w|\le 2$

60. If $\dfrac{dy}{dx}=0\Rightarrow x_0^2+2cx_0-1=0$

So, $x_0y(x_0)=\dfrac{1}{2}\Rightarrow\lambda=\dfrac{1}{2}$

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61. Clearly $\vec{a} \perp \vec{b}; \vec{c} \perp \vec{d}$

And $(\vec{a}+\vec{c}) \perp (\vec{b}+\vec{d})$

62.

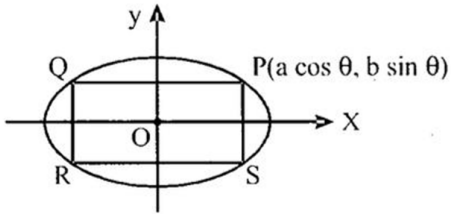
$$\text{Rank} = \frac{4}{2}(6!) + 3\left(\frac{4!}{2}\right) + 2(3!) + 1(2!) + 1! + 0!$$

$$= 1492$$

63. Ans -f $\left(3\frac{\pi}{2}\right) = 2\pi$

$$gl(2\pi) = \frac{1}{fl\left(3\frac{\pi}{2}\right)} = \frac{3}{7}$$

64. Let PQRS be a rectangle, where P is $(a\cos\theta, b\sin\theta)$



$\therefore$ Area of rectangle $\therefore = 4(a\cos\theta),(b\sin\theta) = 2ab\sin 2\theta$

This maximum when $\sin 2\theta = 1$

Hence, maximum area $= 2ab(1) = 2ab$

65. $$\begin{vmatrix} 2 & -1 & -5 \\ 1 & 1 & -1 \\ 3 & 1 & -k \end{vmatrix} = 0$$

$$2(-k-1) + 1(-k+3) - 5(-1-3) = 0$$

$$-3k + 21 = 0$$

K=7

66. $y = \cos\theta\left(\sin\theta + \sqrt{\sin^2\theta + \sin\alpha}\right)$

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$$\frac{y}{\cos\theta}=\sin\theta+\sqrt{\sin^2\theta+\sin^2\alpha}$$

$$y\sec\theta-\sin\theta=\sqrt{\sin^2\theta+\sin^2\alpha}$$

Squaring on both sides

$$y^2\sec^2\theta+\sin^2\theta-2y\tan\theta=\sin^2\theta+\sin^2\alpha$$

$$y^2+y^2\tan^2\theta-2y\tan\theta-\sin^2\alpha=0$$

$$\tan\theta\in R\Rightarrow\Delta\geq 0$$

$$4y^2-4.y^2.\left(y^2-\sin^2\alpha\right)\geq 0$$

$$4y^2\left[1-y^2+\sin^2\alpha\right]\geq 0$$

$$\Rightarrow 1-y^2+\sin^2\alpha\geq 0$$

$$y^2-1-\sin^2\alpha\leq 0$$

$$y^2-\left(1+\sin^2\alpha\right)\leq 0$$

$$y\in\left[-\sqrt{1+\sin^2\alpha},\sqrt{1+\sin^2\alpha}\right]$$

67. $\vec{a}=2\vec{i}-\vec{j}+5\vec{k}\qquad\vec{b}=\alpha\vec{i}+\beta\vec{j}+2\vec{k}$

$$\left(\vec{a}\times\vec{b}\right)=\begin{pmatrix}\hat{i}&\hat{j}&\hat{k}\\2&-1&5\\\alpha&\beta&2\end{pmatrix}=\hat{i}\left(-2-5\beta\right)-\hat{j}\left(4-5\alpha\right)+\hat{k}\left(2\beta+\alpha\right)$$

$$\left(\vec{a}\times\vec{b}\right)\times i=\begin{vmatrix}i&j&k\\-2-5\beta&5\alpha-4&\alpha+2\beta\\1&0&0\end{vmatrix}=(\alpha+2\beta)j+(4-5\alpha)k$$

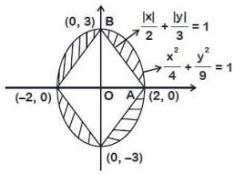
$$[(\vec{a}\times\vec{b})\times\hat{i}]\cdot\hat{k}=\left[(\alpha+2\beta)\hat{j}+(4-5\alpha)\hat{k}\right]\cdot\hat{k}$$

$$\frac{23}{2}=4-5\alpha\qquad\Rightarrow\alpha=\frac{-3}{2}$$

$$\vec{b}\times 2\vec{j}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\\alpha&\beta&2\\0&2&0\end{vmatrix}=-4i+2\alpha k$$

$$|\vec{b}\times 2\hat{j}|=\sqrt{(-4)^2+(2\alpha)^2}=\sqrt{16+4\alpha^2}$$

68.



$\because$ Area of ellipse= $\pi ab=\pi\times 2\times 3=6\pi$

$\because$ Required area =Area of ellipse-4(Area of triangle OPQ)

$=6\pi-4\left(\frac{1}{2}\times 2\times 3\right)=6\pi-12=6\left(\pi-2\right)sq.\,units$ 69.

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$$\int \frac{x^2-1}{x^2\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx = \int \frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx \text{ put } x+\frac{1}{x}=t$$

$$= \int \frac{dt}{t\sqrt{t^2-2}} \quad t=\sqrt{2}\sec\theta \quad = \int \frac{\sqrt{2}\sec\theta\tan\theta d\theta}{\sqrt{2}\sec\theta\sqrt{2}\tan\theta} = \frac{1}{\sqrt{2}}\theta + C$$

70. Conceptual

71. $f(x)=[4x]$ so, discontinuous

at $x=\frac{1}{4},\frac{2}{4},---\frac{15}{4}$

$\Rightarrow$ No.of discontinuous points = 15

72. $$\lim_{x\to 0}\int_0^x \frac{t^2dt}{(x-\sin x)\sqrt{a+t}} = \lim_{x\to 0}\int_0^x \frac{\frac{t^2dt}{\sqrt{a+t}}}{(x-\sin x)} = \lim_{x\to 0}\frac{x^2}{\sqrt{a+x(1-\cos x)}}$$

$$\lim_{x\to 0}\frac{x^2}{2\sin^2\frac{x}{2}.\sqrt{a+x}} = \frac{2}{\sqrt{a}} = 1(given)$$

$\Rightarrow a=4$

73. $\because \sum P(X)=1$

$\therefore k+2k+2k+3k+k=1 \Rightarrow k=\frac{1}{9}$

Now, p= $P\left(\frac{1<X<4}{X<3}\right) = \frac{P(X=2)}{P(X<3)} = \frac{\frac{2k}{9k}}{\frac{k}{9k}+\frac{2k}{9k}} = \frac{2}{3}$

Now, $5p=\lambda k$

$\Rightarrow (5)\left(\frac{2}{3}\right) = \lambda(1/9) \Rightarrow \lambda = 30$

74. Given lines are two skew lines $\Rightarrow$ non coplanar $\Rightarrow$ $\lambda \neq 1$, so minimum $\lambda = 2$

75. For non-trivial solution,

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$$\Delta = \begin{vmatrix} a & \sin b & \cos b \\ 1 & \cos b & \sin b \\ -1 & \sin b & -\cos b \end{vmatrix} = 0$$

$$\Rightarrow a[-\cos^2 b - \sin^2 b] - \sin b[-\cos b + \sin b] + \cos b[\sin b + \cos b] = 0$$

$$\Rightarrow -a + \sin 2b + \cos 2b = 0$$

$$\Rightarrow a = \cos 2b + \sin 2b = \sqrt{2}\cos\left(2b - \frac{\pi}{4}\right)$$

since, $-1 \le \cos\left(2b - \frac{\pi}{4}\right) \le 1$

$$\Rightarrow -\sqrt{2} \le \sqrt{2}\cos\left(2b - \frac{\pi}{4}\right) \le \sqrt{2} \Rightarrow -\sqrt{2} \le a \le \sqrt{2}$$

$$\Rightarrow a \in [-\sqrt{2}, \sqrt{2}]$$

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